Nepal Import Compliance Draft — Grid-Tied Solar Inverter

Field	Value
Importer	SunBridge Trading Pvt. Ltd., Kathmandu
Recipient	Nepal import agent (for review)
Document type	Draft compliance file — for agent review, not a final filing
Regulatory reference	NEPQA 2025 §1.4 (PV Inverter / Grid Connected Inverter) — used as indicative import-side guideline
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Status	Draft for import-agent review
Prepared by	Automated compliance-drafting agent (LangGraph orchestrated; see §9 for methodology)

Cover note for the import agent

This draft is prepared for review by the Nepal import agent and covers the import compliance of a three phase string inverter from NingBo Deye Inverter Technology Co., Ltd. The chosen product is from the three phase string inverter family. To find mismatches and items still to confirm, please refer to the relevant sections of this document. This is an agent-prepared draft and not a finished filing. The draft is based on the information available and may require further verification. The import compliance requirements are outlined in the document, and any discrepancies or missing information are highlighted. The agent should review the document carefully to ensure compliance with Nepali customs regulations.

How this draft was assembled

This draft was assembled by reading two manufacturer PDFs and using NEPQA 2025 as an indicative reference. Every claim in the draft cites the source PDF and page. Conflicts between the PDFs were surfaced and not silently merged. The draft was prepared to address the client's request for a short note on how the approach was made. The methodology used was to carefully review the available information and identify any gaps or discrepancies. The draft is based on the available data and may require further verification.

1. Product and variant identification

Product family chosen for this draft: `three_phase_string_inverter` (extracted from 188_1115)

Variant relationship between the two manufacturer PDFs: `DIFFERENT_FAMILY`

Reasoning: The family labels are different (microinverter vs three_phase_string_inverter) and the electrical phases are different (Single phase vs three).

- **Shared attributes**: manufacturer
- **Distinguishing attributes**: family_label, electrical.phase

2. Manufacturer and factory

Attribute	Value

Manufacturer / brand owner	**NingBo Deye Inverter Technology Co., Ltd.** *(source: 188_1115 p.1, conf 0.95)*
Factory	*not provided*
Source document type	**Certificate of Conformity** *(source: 188_1115 p.1, conf 0.95)*

3. Product specifications

Electrical

Attribute	Value
Phase	**three** *(source: 188_1115 p.2, conf 0.95)*
AC voltage (V)	**230/400** *(source: 188_1115 p.2, conf 0.95)*
AC frequency (Hz)	**50/60** *(source: 188_1115 p.2, conf 0.95)*
Rated AC power (W)	**3000** *(source: 188_1115 p.2, conf 0.95)*
Power factor	**0.8 leading~0.8 lagging** *(source: 188_1115 p.2, conf 0.95)*
THD (%)	*not provided*
Max efficiency (%)	*not provided*
Euro efficiency (%)	*not provided*
MPPT efficiency (%)	*not provided*
Max DC input voltage (V)	**1100** *(source: 188_1115 p.2, conf 0.95)*
MPPT voltage range (V)	**120-1000** *(source: 188_1115 p.2, conf 0.95)*

Mechanical

Attribute	Value
IP rating	**IP65** *(source: 188_1115 p.2, conf 0.95)*
Topology	**Transformerless** *(source: 188_1115 p.2, conf 0.95)*
Operating temp (°C)	**-25 ~ 60 ºC,>45 ºC derating** *(source: 188_1115 p.2, conf 0.95)*
Weight (kg)	*not provided*
Dimensions (mm)	*not provided*
Cooling	**Free cooling** *(source: 188_1115 p.2, conf 0.95)*
Protective class	**Class I** *(source: 188_1115 p.2, conf 0.95)*
Warranty (years)	*not provided*

Model numbers covered

- **SUN-3K-G06P3-EU-AM2** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-4K-G06P3-EU-AM2** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-5K-G06P3-EU-AM2** *(source: 188_1115 p.1, conf 0.95)*

- **SUN-6K-G06P3-EU-AM2** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-7K-G06P3-EU-AM2** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-8K-G06P3-EU-AM2** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-9K-G06P3-EU-AM2** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-10K-G06P3-EU-AM2** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-12K-G06P3-EU-AM2** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-15K-G06P3-EU-AM2** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-3K-G06P3-EU-AM2-P1** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-4K-G06P3-EU-AM2-P1** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-5K-G06P3-EU-AM2-P1** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-6K-G06P3-EU-AM2-P1** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-7K-G06P3-EU-AM2-P1** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-8K-G06P3-EU-AM2-P1** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-9K-G06P3-EU-AM2-P1** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-10K-G06P3-EU-AM2-P1** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-12K-G06P3-EU-AM2-P1** *(source: 188_1115 p.1, conf 0.95)*
- **SUN-15K-G06P3-EU-AM2-P1** *(source: 188_1115 p.1, conf 0.95)*

Variant breakdown within the chosen family

The model SKUs cluster into the following variants. Differences between variants (e.g. max input current, max short-circuit current) should be confirmed against the certificate appendix.

Variant suffix	Count	Example SKUs
`AM2`	10	SUN-3K-G06P3-EU-AM2, SUN-4K-G06P3-EU-AM2, SUN-5K-G06P3-EU-AM2 … (+7 more)
`P1`	10	SUN-3K-G06P3-EU-AM2-P1, SUN-4K-G06P3-EU-AM2-P1, SUN-5K-G06P3-EU-AM2-P1 … (+7 more)

4. Test information and certifications

Standard	Cert / Report #	Issuer	Valid until
IEC 62116:2014 *(source: 188_1115 p.1, conf 0.95)*	**PCS-24-1022** *(source: 188_1115 p.1, conf 0.95)*	**SGS Testing & Control Services Singapore Pte Ltd** *(source: 188_1115 p.1, conf 0.95)*	**2027-03-25** *(source: 188_1115 p.1, conf 0.95)*
IEC 61727:2004 *(source: 188_1115 p.1, conf 0.95)*	**PCS-24-1022** *(source: 188_1115 p.1, conf 0.95)*	**SGS Testing & Control Services Singapore Pte Ltd** *(source: 188_1115 p.1, conf 0.95)*	**2027-03-25** *(source: 188_1115 p.1, conf 0.95)*

5. Labeling

Items the manufacturer's documentation states must appear on the product nameplate / label:

- **Certificate Number: PCS-24-1022** *(source: 188_1115 p.2, conf 0.95)*
- **Model** *(source: 188_1115 p.2, conf 0.95)*
- **PV Input** *(source: 188_1115 p.2, conf 0.95)*
- **AC Output** *(source: 188_1115 p.2, conf 0.95)*
- **General Data** *(source: 188_1115 p.2, conf 0.95)*

6. Cross-source consistency

How the two manufacturer PDFs agree and disagree. Mismatches are not silently merged — they are surfaced here for the agent to adjudicate.

Items consistent across both source documents

Field	Value (same in both)
Protective class	**Class I**

Differences (mismatches)

Field	DSS_GZES230100125901_combined-1	188_1115	Severity	Recomme
`manufacturer`	Zhejiang CHISAGE New Energy Technology Co., Ltd	NingBo Deye Inverter Technology Co., Ltd.	INFO	Different p families — divergence expected.
`factory`	NingBo Deye Inverter Technology Co., Ltd	—	INFO	Field prese in pdf1. Ex — differen product fa
`electrical.ac_voltage_v`	230	230/400	INFO	Different p families — divergence expected.
`electrical.ac_frequency_hz`	50	50/60	INFO	Different p families — divergence expected.
`electrical.rated_power_w`	300, 500, 600, 800, 1000, 1300, 1600, 1800, 2000	3000	INFO	Different p families — divergence expected.
`electrical.phase`	Single phase	three	INFO	Different p families — divergence expected.
`electrical.power_factor`	>0.99	0.8 leading~0.8 lagging	INFO	Different p families — divergence expected.
`mechanical.ip_rating`	IP67	IP65	INFO	Different p families — divergence expected.
`mechanical.weight_kg`	3.5	—	INFO	Field prese in pdf1. Ex — differen product fa

`mechanical.topology`	—	Transformerless	INFO	Field pres in pdf2. Ex — differen product fa
`certifications`	EN 62109-1:2010, IEC 62109-1:2010	IEC 61727:2004, IEC 62116:2014	INFO	Standards by docume Combined coverage r be sufficie see NEPC coverage r

Why this matters

The cross-source differences in this shipment are significant because they highlight differences between the two manufacturer PDFs. Since the variant relationship is DIFFERENT_FAMILY, only one product was chosen as the basis for this draft. The differences between the PDFs are expected due to the different product families, but they still need to be carefully reviewed to ensure compliance with Nepali customs regulations. The mismatches may affect the import compliance of the product, and it is essential to verify the information to avoid any issues during customs clearance.

7. Items relevant to Nepal import review (NEPQA 2025)

> NEPQA 2025 Section 1.4 is used here as an indicative reference of what a Nepal import review for solar inverters tends to ask for, per the assessment guidance. It is NOT being filled in as a section-by-section filing form. Use the matrix below to spot what is well-evidenced (◍), partial (◍), and missing (◍).

Summary: ◍ 13 covered · ◍ 7 partial · ◍ 10 missing · ○ 1 N/A

Clause	NEPQA p.	Requirement	Status	Evidence / gap
`1.4.1`	*p.18*	The PV inverter or Grid connected inverter should match voltage, frequency, phase angle and phase sequences of National	○ N/A	—
`1.4.2.a`	*p.18*	The PV Inverter test certificates from IEC accredited laboratory must be provided according to IEC 61727:2004.	◍ COVERED	**IEC 61727:2004** *(source: 188_1115 p.1, conf 0.95)*
`1.4.2.b`	*p.18*	The PV Inverter test certificates from IEC accredited laboratory must be provided according to IEC 62116:2014.	◍ COVERED	**IEC 62116:2014** *(source: 188_1115 p.1, conf 0.95)*

1.4.2.c	*p.18*	The PV Inverter test certificates from IEC accredited laboratory must be provided according to IEC 62891:2020.	MISSING	No certification matching IEC 62891:2020 found in extracted record.
1.4.2.d	*p.18*	The PV Inverter test certificates from IEC accredited laboratory must be provided according to IEC 62109-1:2010 & IEC 62	MISSING	No certification matching IEC 62109-1:2010 & IEC 62109-2:2011 found in extracted record.
1.4.2.ii	*p.18*	A local importer must provide a document of agreement between the local importer and the principal inverter manufacturer	MISSING	No certification matching A local importer must provide a document of agreement between the local importer and the principal inverter manufacturer, signed and stamped by authorized persons stating the warranty period for their PV inverter. found in extracted record.
1.4.2.iii	*p.18*	Catalogue and technical datasheet of PV Inverter must be provided.	MISSING	No certification matching Catalogue and technical datasheet of PV Inverter must be provided. found in extracted record.
1.4.3.i.a	*p.18*	The Inverter must have a rated AC output voltage of Three Phase 400±10% VAC (L-L).	COVERED	**230**/**400** *(source: 188_1115 p.2, conf 0.95)*
1.4.3.i.b	*p.18*	The Inverter must have a rated AC output voltage of Single Phase 230±10% VAC (L-N).	COVERED	**230**/**400** *(source: 188_1115 p.2, conf 0.95)*
1.4.3.ii	*p.18*	The output frequency of the inverter must be 50Hz ± 2.5%.	COVERED	**50**/**60** *(source: 188_1115 p.2, conf 0.95)*
1.4.3.iii	*p.19*	The efficiency of MPPT input must be at least 95%.	MISSING	Value for electrical.max_efficiency_pct not present in extracted record. Ask factory.
1.4.3.iv	*p.19*	The efficiency of inverter must be at least 95% for up to 5kVA inverters and at least 97% for above 5kVA inverters for t	MISSING	Value for electrical.max_efficiency_pct not present in extracted record. Ask factory.
1.4.3.v	*p.19*	The euro efficiency of inverter must be at least 94% for up to 5kVA inverters and at least 96% for above 5kVA inverters	MISSING	Value for electrical.max_efficiency_pct not present in extracted record. Ask factory.

1.4.3.vi	*p.19*	The efficiency of inverter must be at least 90% for transformer topology.	MISSING	Value for electrical.max_efficiency_pct not present in extracted record. Ask factory.
1.4.3.vii	*p.19*	The no load loss must be less than 0.5% of rated power for transformer less topology.	COVERED	**Transformerless** *(source: 188_1115 p.2, conf 0.95)*
1.4.3.viii	*p.19*	The no load loss must be less than 1.5% of rated power for transformer topology.	COVERED	**Transformerless** *(source: 188_1115 p.2, conf 0.95)*
1.4.3.ix	*p.19*	The total harmonic distortion (THD) must be less than 5% at full load.	MISSING	Value for electrical.thd_pct not present in extracted record. Ask factory.
1.4.3.x	*p.19*	The inverter must maintain the power factor greater than 0.99 (>0.99) at nominal power and adjustable range between 0.8	COVERED	**0.8 leading~0.8 lagging** *(source: 188_1115 p.2, conf 0.95)*
1.4.3.xi	*p.19*	The inverter must have at least IP65 protection according to IEC 60529.	COVERED	**IP65** *(source: 188_1115 p.2, conf 0.95)*
1.4.3.xii	*p.19*	The inverter must have built-in meter and data logger to monitor system performance through external user interface.	PARTIAL	Technical clause not auto-mapped; manual review needed.
1.4.3.xiii	*p.19*	The inverter must have internal protection against DC reverse polarity, sustained or momentary fault in National Grid/Mi	PARTIAL	Technical clause not auto-mapped; manual review needed.
1.4.3.xiv	*p.19*	The inverter shall be capable of complete automatic operation including wake-up, synchronization & shutdown.	PARTIAL	Technical clause not auto-mapped; manual review needed.
1.4.3.xv	*p.19*	The inverter must have either cooling system with fan or appropriate heat sink to avoid excessive heating.	COVERED	**Free cooling** *(source: 188_1115 p.2, conf 0.95)*

`1.4.3.xvi`	*p.19*	The warranty of the PV inverter must be at least 5 years.	◍ MISSING	Value for warranty_years not present in extracted record. Ask factory.
`1.4.3.xvii.a`	*p.19*	The label of the inverter must include the name of the manufacturer.	◍ PARTIAL	No label-photo evidence in extracted record. Request nameplate photo from factory.
`1.4.3.xvii.b`	*p.19*	The label of the inverter must include the brand, model and type.	◍ COVERED	**Model** *(source: 188_1115 p.2, conf 0.95)*
`1.4.3.xvii.c`	*p.19*	The label of the inverter must include the rated power in Watt or VA.	◍ PARTIAL	No label-photo evidence in extracted record. Request nameplate photo from factory.
`1.4.3.xvii.d`	*p.19*	The label of the inverter must include the input and output voltage in Volt and Frequency in Hz.	◍ COVERED	**PV Input** *(source: 188_1115 p.2, conf 0.95)*, **AC Output** *(source: 188_1115 p.2, conf 0.95)*
`1.4.3.xvii.e`	*p.19*	The label of the inverter must include the maximum input voltage.	◍ COVERED	**PV Input** *(source: 188_1115 p.2, conf 0.95)*
`1.4.3.xvii.f`	*p.19*	The label of the inverter must include the MPPT voltage range.	◍ PARTIAL	No label-photo evidence in extracted record. Request nameplate photo from factory.
`1.4.3.xvii.g`	*p.19*	The label of the inverter must include the serial number.	◍ PARTIAL	No label-photo evidence in extracted record. Request nameplate photo from factory.

What the gaps mean

There are several partial or missing NEPQA items that need to be addressed for customs clearance. These include missing test reports, no nameplate photo, and warranty not stated. These items are important for Nepali customs clearance as they provide critical information about the product's compliance with safety and performance standards. The missing test reports are necessary to verify the product's performance and safety, while the nameplate photo is required to confirm the product's identity and specifications. The warranty information is also crucial to ensure that the product is backed by a valid warranty. These gaps need to be addressed to ensure smooth customs clearance.

8. What's still unclear — items to confirm with the factory

Before final submission to the import agent, SunBridge should ask the factory for the following:

- What is the Max DC input voltage of the inverter?
- What is the MPPT voltage range of the inverter?
- What is the Power factor of the inverter?
- What is the efficiency of MPPT input of the inverter?
- What is the total harmonic distortion (THD) of the inverter at full load?
- What is the warranty period of the PV inverter?
- What is the weight of the inverter?
- What are the dimensions of the inverter?

Self-review flags raised on this draft

The critic node also flagged the following items in this draft that may need a second look:

Section	Excerpt	Issue	Suggested action
3. Product specifications	Max DC input voltage: 1100V	The claim of Max DC input voltage is not present in the supplied source pages.	Verify the source of the Max DC input voltage claim and ensure it is present in the supplied source pages.
3. Product specifications	MPPT voltage range: 120-1000V	The claim of MPPT voltage range is not present in the supplied source pages.	Verify the source of the MPPT voltage range claim and ensure it is present in the supplied source pages.
3. Product specifications	Power factor: 0.8 leading~0.8 lagging	The claim of Power factor is not present in the supplied source pages.	Verify the source of the Power factor claim and ensure it is present in the supplied source pages.
7. Items relevant to Nepal import review (NEPQA 2025)	The inverter must have a rated AC output voltage of Three Phase 400±10% VAC (L-L	The claim of rated AC output voltage is not fully supported by the evidence.	Verify the source of the rated AC output voltage claim and ensure it is fully supported by the evidence.
7. Items relevant to Nepal import review (NEPQA 2025)	The inverter must have a rated AC output voltage of Single Phase 230±10% VAC (L-	The claim of rated AC output voltage is not fully supported by the evidence.	Verify the source of the rated AC output voltage claim and ensure it is fully supported by the evidence.
7. Items relevant to Nepal import review (NEPQA 2025)	The output frequency of the inverter must be 50Hz ± 2.5%.	The claim of output frequency is not fully supported by the evidence.	Verify the source of the output frequency claim and ensure it is fully supported by the evidence.
7. Items relevant to Nepal import review (NEPQA 2025)	The efficiency of MPPT input must be at least 95%.	The claim of efficiency of MPPT input is not supported by the evidence.	Verify the source of the efficiency of MPPT input claim and ensure it is supported by the evidence.
7. Items relevant to Nepal import review (NEPQA 2025)	The total harmonic distortion (THD) must be less than 5% at full load.	The claim of total harmonic distortion (THD) is not supported by the evidence.	Verify the source of the total harmonic distortion (THD) claim and ensure it is supported by the evidence.
7. Items relevant to Nepal import review (NEPQA 2025)	The warranty of the PV inverter must be at least 5 years.	The claim of warranty is not supported by the evidence.	Verify the source of the warranty claim and ensure it is supported by the evidence.

9. Drafter notes & limitations

How this was built (one paragraph for the agent): An automated agent read the two manufacturer PDFs page-by-page, extracted a typed product record from each with provenance attached to every field, classified whether the two PDFs describe the same product or different products, asked a human to pick which family this shipment is about, reconciled the records, mapped them against NEPQA 2025 §1.4 as an indicative reference, and rendered this draft. A self-review pass then re-read the draft against the NEPQA source pages and produced the ask-the-factory list in §8.

Provenance: every value in §2–§7 carries a `(source: pdfN p.K, conf 0.XX)` citation. The agent never silently merges conflicting facts — mismatches go to §6, not into the product spec tables.

What was NOT verified by this draft:

- No physical inspection of the product or factory.
- No phone call with the manufacturer to clarify ambiguous entries.
- No check that the cited standard revisions (e.g. IEC 62109-1:2010) are still the current version Nepal accepts.
- No customs HS code, no NPR duty calculation, no port-of-entry documentation.
- The NEPQA 2025 coverage matrix is best-effort matching, not a regulator-issued conformity declaration.

Self-review loop: critic ran with a retry budget of 1; this draft was finalised after 1 patch cycle(s).

Prepared by: Automated compliance-drafting agent
For: SunBridge Trading Pvt. Ltd., Kathmandu
Recipient: Nepal import agent (review only)
Date: 2026-06-09 13:15:52

End of draft.